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Assessing Factors Associated With Patient-Provider Discussions About Lung Cancer Screening Using Andersen's Behavioral Model of Health Services Utilization

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Abstract

Background: Patient-provider shared decision-making discussions are an important component of lung cancer screening guidelines, but little is known about factors associated with these discussions among screening-eligible patients. We used Andersen's Behavioral Model of Health Services Utilization to examine factors associated with discussing LCS with a provider.

Patients: Data came from an online survey of $N = 516$ U.S. adults meeting United States Preventive Services Task Force LCS eligibility criteria (ie, were 50-80 years of age and had at least a 20-pack year history of tobacco use).

Methods: We used logistic regression to investigate whether having a LCS discussion was associated with predisposing factors (eg, chronic obstructive pulmonary disease [COPD] diagnosis), enabling factors (eg, having a primary care provider [PCP]), and need factors (eg, smoking history).

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Disclosure

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CRediT authorship contribution statement

Owen Y. Cai: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Jessica R. Fernandez:** Writing – review & editing, Validation, Methodology, Conceptualization. **Melinda C. Aldrich:** Writing – review & editing, Supervision, Conceptualization. **Jennifer Richmond:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization.

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Results: About 36% participants had ever discussed LCS with a provider. Participants diagnosed with COPD (OR = 3.56, 95% CI, 1.90-6.67) and who had a first-degree relative with lung cancer (OR = 1.78, 95% CI, 1.02-3.09) had higher odds of LCS discussion than those without COPD and with no family history, respectively. Women had lower odds of LCS discussion than men (OR = 0.37, 95% CI, 0.24-0.58). Participants with an income of $30,000 to $59,999 had higher odds of LCS discussion compared to those earning < $30,000 (OR = 1.78, 95% CI, 1.06-2.98). Not having a PCP (OR = 0.39, 95% CI, 0.21-0.72) and currently smoking (OR = 0.38, 95% CI, 0.18-0.79) were associated with lower odds of LCS discussion.

Conclusions: Future interventions are needed to ensure all LCS-eligible individuals have access to provider discussions about LCS.

Keywords

Early detection; Equity; Health services research; Social determinants; Communication

Introduction

Lung cancer is the leading cause of cancer death for men and women in the U.S., with an estimated 234,580 new cases and 125,070 new deaths in 2024.[1] Many lung cancers are detected at later stages when the disease has spread to other sites, resulting in a high mortality rate and low 5-year relative survival rate (6.6% for males, 9.6% for females) for distant lung cancer.[2] Lung cancer screening (LCS) using low-dose computed tomography (LDCT) can help detect lung cancer at earlier, more treatable stages. The National Lung Screening Trial found that LDCT significantly reduced lung cancer mortality compared to chest radiography.[3] Based on evidence from the National Lung Screening Trial and other recent studies,[4] the United States Preventive Services Task Force (USPSTF) currently recommends annual LDCT for eligible individuals. Per USPSTF guidelines, individuals are eligible for LDCT if they are between 50 and 80 years old, have at least a 20-pack year smoking history, and currently smoke or have quit within the last 15 years.[5] Most public and private insurers now cover LDCT for eligible individuals, in part because the Affordable Care Act established insurance coverage requirements for preventive services recommended by the USPSTF.[6] Despite defined eligibility criteria and insurance coverage, LDCT uptake remains low in the U.S. Studies report LDCT national rates ranging from 1.9% in 2016 to 16.4% in 2022, although it is important to note the expansion of guidelines in 2021.[7-10]

There are many potential factors contributing to low LDCT uptake (eg, limited screening awareness, patient fear about a potential cancer diagnosis, and long travel distances to a screening center).[11,12] Limited patient-provider discussions about LCS are another important, but understudied, factor that may contribute to low LCS uptake. Currently, patient-provider shared decision-making discussions are required before receiving LDCT under Centers for Medicare and Medicaid Services guidelines and recommended in USPSTF guidelines.[5,13] These discussions are an important opportunity for patients and providers to weigh potential screening benefits (eg, detecting lung cancer early) with potential harms (eg, false-positive results).[14] Receiving a health care provider's recommendation is one of the strongest predictors of screening uptake for any cancer screening.[15,16] Similar patterns are emerging with LCS. For example, in a prospective observational study of 137 patients

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eligible for LCS, 88.3% of participants moved forward with LCS after patient-centered shared decision-making in an in-person and telephone format.[17] In a smaller survey of 30 patients, half strongly believed that a physician recommendation was the most influential factor to pursue a screening.[18]

The limited research examining the prevalence of patient-provider LCS discussions suggests the prevalence of these discussions is low. Studies on specific cycles of the Health Information National Trends Survey (HINTS) from 2014 to 2022 have found that for individuals estimated to have LCS eligibility, a range of 9.4% to 18% had discussed LCS with their provider in the past year.[19-22] Prior research on this topic has largely used data from nationally representative datasets (eg, HINTS), which typically do not measure detailed pack-year smoking history information for survey respondents. Therefore, prior studies have evaluated LCS discussions among patients estimated to be eligible for LDCT (eg, those who meet LDCT age-eligibility requirements and have any history of smoking). Accordingly, more research is needed to go beyond estimations and understand patient-provider discussions about LCS among patients meeting LDCT screening eligibility criteria. We aimed to address this knowledge gap by examining factors associated with discussing LCS with a provider among patients meeting USPSTF eligibility criteria for LCS.

Methods

Conceptual Framework

This study was guided by Andersen's Behavioral Model of Health Services Utilization. Andersen's Behavioral Model posits that factors impacting the use of a health service (eg, LCS) can be broadly grouped into 3 categories: predisposing, enabling, and need factors.[23] Predisposing factors include demographic factors, social structures, and environments that exist before the need for a health service arises, and these predisposing factors affect people's baseline need and capacity to access a service.[24] Enabling factors comprise the necessary resources an individual must have to access health services, such as transportation or health insurance. Need factors directly prompt or cause people to seek out a health service, such as specific health symptoms or being eligible for a health service (eg, LCS). We hypothesized that enabling factors (eg, income, health insurance) would be significantly associated with odds of ever having a LCS discussion with a provider. Our hypothesis focused on enabling factors because these factors may impact the accessibility of LCS services and discussions once a patient's need and eligibility for screening has been established.

Study Population and Recruitment

We used data from a survey of U.S. adults recruited through CloudResearch's Prime Panels from July-September 2022.[25] Prime Panels has been shown to facilitate recruitment of diverse participant samples.[26] Participants were eligible to complete the online survey if they met USPSTF eligibility criteria for LCS (ie, were 50-80 years of age, had at least a 20-pack year history of tobacco use, and currently smoked or had quit smoking within the past 15 years). We used survey questions informed by prior research to calculate pack-year smoking history and multiply the number of years smoked by the average number of packs smoked

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per day.[27,28] Participants needed to reside in the U.S., be comfortable reading and talking in English, and not have a prior diagnosis of lung cancer. Potential participants completed an eligibility survey in REDCap,[29] and eligible participants were able to complete the main study survey. We formatted the eligibility survey so that it was accessible on mobile devices, tablets, and computers. Participants were recruited so that about half of the sample self-identified as Black or African American, due to the disproportionately higher lung cancer burden and lower LCS rates in Black communities.[30-32] In addition, we aimed to prioritize representation from participants across varying levels of education and income. The Vanderbilt University Medical Center Institutional Review Board approved this study, and all participants provided informed consent.

Study Measures and Variables

Variables selected for inclusion in analyses were based on prior literature.[19,21,22,33] As described below, we then grouped these variables into 3 categories based on Andersen's Behavioral Model of Health Services Utilization (ie, predisposing, enabling, and need factors).

Outcome variable. Participants were asked "Has a doctor or health care provider ever talked to you about getting a test to look for early lung cancer?"[28] Participants could answer yes or no.

Enabling variables. We included health insurance status (insured vs. not insured), income (less than $30,000; $30,000 to $59,999; and more than $60,000), employment status (employed, unemployed, or retired), and lack of a primary care provider (PCP) as enabling factors.

Need variables. We included age, pack-year smoking history, and current smoking status as need variables because these factors comprise the USPSTF eligibility criteria for LCS. Current smoking status was measured dichotomously (currently smoke vs. formerly smoked).

Predisposing variables. We included prior chronic obstructive pulmonary disease (COPD) diagnosis, number of comorbidities, and family history of lung cancer as predisposing factors given their association with lung cancer risk.[34-36] Number of comorbidity diagnoses (eg, high blood pressure and heart disease) was measured continuously using the Self-Administered Comorbidity Questionnaire.[37] Family history of lung cancer was assessed by asking whether participants had a first degree relative diagnosed with lung cancer, a second degree relative diagnosed with lung cancer, or no family history of lung cancer.[38] We also included U.S. census region, race, ethnicity, education level, and marital status as additional predisposing demographic variables.

Statistical Analysis

We used descriptive statistics to summarize participant characteristics. We conducted univariate logistic regression to examine unadjusted associations between each of the predisposing, enabling, and need factors and odds of LCS discussion. We then used

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complete case multivariable logistic regression to evaluate associations between the predisposing, enabling, and need factors and odds of LCS discussion. For both unadjusted and adjusted models, we calculated odds ratios, 95% confidence intervals, and *P*-values. We used 4.2.2 and RStudio Version 2023.12.0 for data analyses.[39]

Results

Participant Characteristics

A total of $N = 516$ participants completed the survey. The mean age of participants was 61.3 years (SD = 7.2 years). Approximately half of participants identified as women (50%), Black (48%), and resided in the U.S. South (47%). About one-third of participants had a high school education or lower (29%) and were currently employed (34%). About 42% of participants had annual household incomes below $30,000. Most participants currently had health insurance (92%), no family history of lung cancer (70%), and currently smoked (89%). On average, participants had a pack-year smoking history of 42.4 years (SD = 25.7 years). About 17% of participants had a prior diagnosis of COPD. Just over a third of participants (36%) had ever discussed LCS with a health care provider, and roughly 81% had a PCP. A total of $N = 352$ (68%) had ever received LCS. Table 1 presents a full breakdown of participant characteristics.

Unadjusted Logistic Regression

Table 2 presents results from the unadjusted logistic regression models.

Enabling factors: Income and having a PCP were positively associated with odds of LCS discussion. Those with an income of $30,000 to $59,999 had increased odds of LCS discussion relative to those with an income of less than $30,000 (OR = 1.66, 95% CI, 1.09-2.54). The same held true for those with incomes greater than $60,000 compared to those with less than $30,000 (OR = 1.66, 95% CI, 1.05-2.65). Patients without a PCP had lower odds of LCS discussion compared to patients with a PCP (OR = 0.41, 95% CI, 0.24-0.70). The other enabling factors, employment status and health insurance, were not associated with odds of LCS discussion.

Need factors: None of the need factors were significantly associated with LCS discussion odds.

Predisposing factors: COPD diagnosis and gender were associated with odds of LCS discussion. Participants diagnosed with COPD had greater odds of LCS discussion compared to those not diagnosed with COPD (OR = 2.62, 95% CI, 1.63-4.19). Identifying as a woman (vs. a man) was associated with lower odds of LCS discussion (OR = 0.44, 95% CI, 0.30-0.63). Number of chronic diseases, family history of lung cancer, race/ethnicity, education, marital status, and census region were not associated with odds of LCS discussion.

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Adjusted Logistic Regression

Table 3 presents adjusted logistic regression results. The analytic sample for the adjusted logistic regression included $N = 481$ participants after accounting for $n = 35$ participants with missing data.

Enabling factors: Participants with an income of $30,000 to $59,999 still had higher odds of LCS discussion compared to those with an income less than $30,000 (OR = 1.78, 95% CI, 1.06-2.98), although having an income greater than $60,000 was not significantly associated with LCS discussion in the adjusted model. Lack of a PCP remained associated with lower odds of LCS discussion (OR = 0.39, 95% CI, 0.21-0.72). Employment and health insurance status were not associated with odds of having a LCS discussion.

Need factors: Current smoking status was associated with odds of LCS discussion. Participants who currently smoked had lower odds of LCS discussion than those who formerly smoked (OR = 0.38, 95% CI, 0.18-0.79). Age and pack-year smoking history were not associated with odds of LCS discussion.

Predisposing factors: COPD diagnosis and gender remained significantly associated with odds of LCS discussion in the adjusted model. Participants diagnosed with COPD had higher odds of LCS discussion than those without COPD (OR = 3.56, 95% CI, 1.90-6.67), and women had lower odds of LCS discussion than men (OR = 0.37, 95% CI, 0.24-0.58). In addition, patients who had a first-degree relative diagnosed with lung cancer had higher odds of LCS discussion compared to those with no family history of lung cancer (OR = 1.78, 95% CI, 1.02-3.09). All other predisposing factors were not associated with odds of LCS discussion.

Discussion

In this study, we found that only 36% of participants had ever discussed LCS with a provider —a low percentage that is particularly concerning given that all participants in this study were eligible for LCS. This low percentage also aligns with previous studies, which have found the prevalence of patient-provider LCS discussions to be under 20%.[19-22,40] To our knowledge, this study is one of the first to assess LCS discussions among patients eligible for screening rather than using secondary datasets to estimate patient screening eligibility. Although it is somewhat encouraging that a slightly greater percentage patients in our study than those in previous studies had ever discussed LCS with a provider (36% vs. < 20%), the overall low prevalence of screening discussions suggests many screening-eligible individuals may be missing out on life-saving opportunities to receive LCS.[20,41]

Determinants of LCS patient-provider discussions are scarcely researched, and our study adds to the literature by investigating enabling, need, and predisposing factors associated with odds of discussing LCS with a health care provider. Below, we discuss the salience of our findings across enabling, need, and predisposing factors.

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Enabling Factors

We hypothesized that enabling factors would be significantly associated with our LCS discussions outcome because these factors (eg, having a PCP) could plausibly make it easier or harder to access screening among eligible patients. We found that not having a PCP was associated with lower odds of LCS discussion, in line with previous research.[22] Indeed, prior research suggests consistent access to a PCP is associated with positive health outcomes broadly, including better self-reported health and uptake of preventive health services.[42-44] More specifically, receiving a recommendation from a provider is a strong predictor of patient receipt of cancer screening for various cancer types.[15,16] Having a PCP could be associated with higher odds of LCS discussion because PCPs may consistently assess a patient's relevant medical history, understand what health services are appropriate, and approach conversations with a tailored communication style. The inverse may also hold, as not having a PCP may place the onus on a patient to seek LCS and explain their relevant medical history, creating the opportunity for missed screenings.

We also found that income was associated with LCS discussions. Screening costs are consistently a concern among LCS-eligible patients, and these concerns likely pose a greater burden for patients with low incomes.[11,45] It is also possible that patients anticipate downstream financial costs in the case of a positive screening result, and prefer to avoid these potential costs. Of note, our adjusted model found that participants with an income of $30,000 to $59,999 had higher odds of LCS discussion compared to those with less than $30,000, but having an income greater than $60,000 was not significantly associated with LCS discussions. It is possible that having a higher income facilitates LCS for those earning $30,000 to $59,999, but this marginal benefit diminishes with incomes greater than $60,000. It is also possible that factors other than economic concerns play a larger role in facilitating LCS discussions for patients with incomes greater than $60,000, but future research with a larger sample size is needed to evaluate these hypotheses. Future system-level interventions are also needed to increase access to enabling resources that may promote LCS discussions (eg, access to PCPs and higher incomes).

We also found that employment and health insurance status were not associated with odds of LCS discussions. These findings were counter to our hypotheses because health insurance facilitates access to LCS in the U.S., and employment status could affect patients' ability to take time off work to visit a provider. However, prior literature has also found that health insurance status was not associated with LCS discussions.[19,22] It is possible that enabling factors like employment and health insurance status affect patient odds of completing a LCS test but not the odds of discussing the procedure with a provider.

Need Factors

Smoking status was significantly associated with having had a LCS discussion. Participants that currently smoked were less likely than participants who formerly smoked to have discussed LCS with a provider. This finding may initially seem counterintuitive, given that health care providers often inquire about current smoking status during medical appointments, and learning that a patient currently smokes might serve as a key reminder to discuss LCS eligibility.[46,47] However, smoking and lung cancer are heavily stigmatized

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by society and health care providers, potentially contributing to biases and inaccurate assumptions that individuals who currently smoke do not care about their health or are not interested in screening.[47-52] Such assumptions may be compounded by prior research suggesting that individuals who currently smoke are less likely than those who formerly smoked to get screened for other cancer types, such as colorectal, breast, and prostate.[53] Patients who currently smoke may worry that initiating LCS discussions may cause providers to shame them for their smoking behaviors. Patients may also believe that they do not deserve an early detection service if they actively smoke, which might be a barrier to initiating a LCS discussion.[54] Moreover, future research should consider how prior attempts to quit smoking factor into the likelihood of a LCS discussion. Based on prior qualitative interviews, some patients that currently smoke may believe screening reduces the importance of smoking cessation.[55] Providers who are wary of these beliefs may not initiate a LCS discussion for those who currently smoke and have not attempted to quit smoking, although this hypothesis needs empirical evaluation. Overall, future research and interventions are needed to reduce smoking-related stigma and promote patient-provider discussions among screening-eligible patients who currently smoke. It is particularly important that these LCS discussions occur in a manner that addresses LCS misconceptions and avoids further stigmatization.

The other need variables, age and pack-year smoking history, were not associated with odds of LCS discussion. One interpretation is that among age-eligible patients, having a pack-year smoking history beyond the 20 pack-year threshold does not significantly affect the odds of having an LCS discussion, potentially because everyone beyond this smoking threshold is screening-eligible. Importantly, our entire sample consisted of patients who met USPSTF eligibility criteria for LCS in terms of their age and smoking history. Therefore, it is also possible that age and smoking history were not associated with our outcome because everyone in the sample was screening-eligible and had a similar need for LCS. Additionally, our sampling approach limits our ability to compare odds of LCS discussions between eligible and noneligible patients, though noneligible patients may not need to discuss LCS with their provider.

Predisposing Factors

We identified 3 predisposing factors—COPD diagnosis, family history of lung cancer, and gender— associated with LCS discussions. The odds of having discussed LCS with a provider were about 3.5 times greater for participants diagnosed with COPD compared to those not diagnosed with COPD. COPD is a strong risk factor for developing lung cancer and patients and their providers may be primed to discuss lung health and screening options if the patient has this condition.[35,56] Individuals with COPD may have greater familiarity with lung cancer as a result of their medical history, which may also increase their likelihood of discussing screening with their provider as part of ongoing conversations about their lung health. The validated Self-Administered Comorbidity Questionnaire was used to control for the impact of other comorbidities in our model.[37] As an aggregated measure of comorbidities, it revealed no significant association with LCS discussion. Future research may benefit from using a disaggregated measure to examine the impact of specific comorbidities (eg, heart disease) on the odds of LCS discussions.

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Having a first degree relative diagnosed with lung cancer was also associated with higher odds of LCS discussions. This family history of lung cancer may similarly prompt patients and health care providers to discuss LCS due increased genetic risk.[36,57] We also found that women had lower odds of having discussed LCS with a provider than men. Prior research on this topic has yielded inconsistent results, with some studies finding no association between sex or gender and LCS discussions, but others similarly finding that women were less likely than men to have a LCS discussion.[19,22,58] These inconsistencies may stem from the different methods used and populations sampled. It is also possible that providers perceive women as having a lower lung cancer risk due to epidemiological trends and cultural perceptions.[59-62] Although the prevalence of smoking among men and women has recently trended towards similar rates, men have historically smoked at greater rates.[61,63] Moreover, successful tobacco campaigns in the 20th century centered heavily on creating a masculine appeal for cigarettes.[60,62] The strong causal relationship between smoking and lung cancer and the fact that men have historically smoked more than women could facilitate biased perceptions among providers that women have a reduced need for LCS. Further research is needed to assess how and why gender may affect patient and provider predispositions for discussing LCS.

Race/ethnicity, education, marital status, and census region were not significantly associated with odds of LCS in our study. Importantly, these results do not necessarily mean that there are no differences in LCS discussions based on these factors as our sample size and sampling approach may have influenced our findings. Further investigation of potential disparities in LCS discussions across predisposing factors is needed. It is also vital that future interventions, policies, and research prioritize health equity and regularly evaluate program outcomes to ensure inequities across predisposing factors do not develop or widen over time.

Implications

Collectively, our findings suggest novel interventions and policies are necessary to support LCS discussions between screening-eligible patients and their providers. Multilevel interventions are important to consider. For example, interventions might focus on increasing LCS knowledge at the patient and provider levels, incorporating screening reminders in electronic health records at the organizational levels, and improving screening access and reducing costs at the policy levels.

First, tailored outreach to women, patients who currently smoke, individuals without a PCP, and communities with lower incomes may be particularly important to ensure equitable access to LCS. Organizations should consider conducting more community outreach events for LCS and disease awareness. Through in-person canvassing or social media, their messaging could highlight the importance of LCS discussions and tailor outreach to underserved patients. Outreach communication might consider focusing more on age and smoking status rather than the calculation for pack-year smoking history to increase LCS message clarity and encourage patients to have more specific conversations about pack-year smoking history with their providers. It is also important to regularly evaluate these efforts, which might involve tracking the quantity of people reached and number of individuals that

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make contact with screening centers or services. As an increasing number of people utilize the internet and social media in their daily lives, tailored advertising campaigns have also increased engagement with online educational materials and the number of scheduled LCS appointments.[64] Since a large number of LCS-eligible individuals are older and potentially less familiar with online platforms, advertising campaigns might consider focusing on younger family members and caregivers in addition to older screening-eligible patients.

Interventions tailored for health care providers also represent an important opportunity to increase patient-provider LCS discussions. Provider knowledge of LCS eligibility criteria and ability to recognize those who are underserved and under-screened is crucial. Indeed, providers who better understand LCS and its eligibility criteria have higher odds of making appropriate referrals.[65] Continuing education courses provide another intervention opportunity to build more specific and empirical provider knowledge. Our findings may also point to the importance of encouraging providers who see high numbers of patients without PCPs (eg, providers in urgent care settings, walk-in clinics, community health centers) to engage their patients with tailored LCS decision tools to help facilitate discussions. This approach may be especially useful to help equitably identify and initiate screening discussions with women, patients who currently smoke, patients with lower incomes, and other patients that may be less likely to have a LCS discussion. Regular evaluation of any interventions implemented (eg, via surveys to assess knowledge/awareness and by examining electronic health records to track rates of LCS receipt over time) is also important to help identify opportunities for quality improvement.

Expanding patient and provider knowledge of web-based decision aids or virtual telehealth services may also increase access to necessary LCS discussions.[66] Those who face transportation or other barriers to visiting a doctor in-person may benefit from the ease of accessing provider expertise virtually. Integrating LCS with other well-established preventive health services may also represent a novel point of intervention. For example, recent campaigns and interventions have promoted LCS alongside breast cancer screening in provider offices, leveraging a more well-known cancer screening type among women and the general population.[67-69] Interventionists could also explore how this approach could be expanded to pair LCS discussions with conversations about other screenings, such as prostate cancer screening. However, for all of the aforementioned interventions, costs, benefits, and equity impacts must be carefully considered.

Limitations

Our study has important limitations. We conducted a crosssectional study that used nonprobability sampling to survey patients eligible for LCS. We also recruited participants using an online panel, and participants without internet access may have been excluded. Although online panels can yield valid and reliable data from diverse samples[70-72] and our study sample mirrors demographics of the U.S. screening-eligible population,[73] our results may not be generalizable to the population of screening-eligible individuals in the U.S. Moreover, the survey did not ask about the depth or quality of the patient-provider discussion about screening or whether the patient or their health care provider initiated the screening discussion, which may represent important context for future research to

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consider. Despite these limitations, our study surveyed a diverse sample of screening-eligible individuals throughout the U.S. and advances the scientific understanding of factors associated with LCS discussions among screening-eligible individuals. Additionally, our study suggests it may be feasible to leverage online panels to recruit larger, nationally representative samples of LCS screening-eligible individuals to replicate our findings or test novel hypotheses.

Conclusion

In this study examining predisposing, enabling, and need factors associated with LCS discussions, only 36% of participants had ever discussed LCS with a provider. We found that having higher income, being diagnosed with COPD, and having at least one first degree relative diagnosed with lung cancer were associated with higher odds of LCS discussions. Conversely, women, individuals without a PCP, and those who currently smoked had lower odds of having a LCS discussion. Future interventions and policies are needed to ensure all LCS-eligible individuals have access to provider discussions about LCS. Such discussions are a precursor to a potentially lifesaving approach to the early detection of lung cancer.

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Clinical Practice Points

- Patient-provider discussions about lung cancer screening (LCS) are critical for helping patients make informed decisions, but there is a paucity of research examining factors associated with having these discussions among patients who are LCS-eligible.
- In this survey of patients eligible for lung cancer screening ($N = 516$), only 36% had ever discussed screening with a provider.
- Certain enabling, need, and predisposing factors (eg, income, gender, COPD diagnosis, family history, smoking status, and lack of a primary care provider) were associated with odds of having screening discussions.
- Multilevel interventions are needed to ensure patients without a primary care provider, with low income, who are female, and who currently smoke have equitable access to patient-provider discussions about LCS.

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Table 1

Participant Characteristics, $N = 516$

Characteristic	N (%) or Mean (SD)
Age	
Mean (SD)	61.3 (7.3)
Range	50-80
Gender	
Man	256 (49.6%)
Woman	253 (49%)
Missing	7 (1.4%)
Race and Ethnicity	
White	249 (48.3%)
Black/African American	246 (47.7%)
Asian/Hispanic/Latino/Multiple Races or Ethnicities	20 (3.9%)
Missing	1 (0.2%)
Highest educational attainment	
High school equivalent or less	149 (28.9%)
Some college	247 (47.9%)
Bachelor's degree	85 (16.5%)
Postgraduate degree	31 (6%)
Missing	4 (0.8%)
Income	
Less than $30,000	214 (41.5%)
$30,000-$59,999	174 (33.7%)
$60,000 or More	124 (24%)
Missing	4 (0.8%)
Employment status	
Employed	176 (34.1%)
Unemployed/other	147 (28.5%)
Retired	193 (37.4%)
Marital status	
Never married	105 (20.3%)
Married	216 (41.9%)
No longer with spouse	192 (37.2%)
Missing	3 (0.6%)
U.S. Census region	
South	242 (46.9%)
Northeast	89 (17.2%)
Midwest	121 (23.5%)
West	64 (12.4%)

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Characteristic	N (%) or Mean (SD)
Health insurance coverage	
Has health insurance	473 (91.7%)
No insurance	41 (7.9%)
Missing	2 (0.4%)
Primary care provider	
No	98 (19%)
Yes	417 (80.8%)
Missing	1 (0.2%)
Prior COPD diagnosis	
No	428 (82.9%)
Yes	88 (17.1%)
Number of chronic diseases	
Mean (SD)	2.484 (2.2)
Range	0-11
Family history of lung cancer	
No family history of lung cancer or unsure	359 (69.6%)
At least one first degree relative had lung cancer	81 (15.7%)
At least one second degree relative had lung cancer	74 (14.3%)
Missing	2 (0.4%)
Pack-year smoking history	
Mean (SD)	42.4 (25.7)
Range	19.5 - 288.0
Smoking status	
Currently smoke	460 (89.1%)
Formerly smoked	56 (10.9%)
Ever discussed lung screening with provider	
No	318 (62.0%)
Yes	187 (36.0%)
Missing	11 (2%)
Ever received lung cancer screening	
Yes	352 (68.3)
No	133 (25.8)
Don’t know	30 (5.8)

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Table 2

Unadjusted Associations Between Predisposing, Enabling, and Need Factors and Odds of Lung Cancer Screening Discussion

Variable Name	OR	95% CI	*P*-Value
Enabling			
Health insurance status			
Ref: has insurance	-	-	-
No insurance	0.52	(0.25-1.09)	.08
Income			
Ref: less than $30,000	-	-	-
$30,000-$59,999	1.66	(1.09-2.54)	.02
$60,000 or more	1.66	(1.05-2.65)	.03
Employment status			
Ref: employed	-	-	-
Unemployed/other	0.68	(0.43-1.08)	.10
Retired	0.92	(0.60-1.40)	.69
Primary care provider (PCP) status			
Ref: has a PCP	-	-	-
Lack of a PCP	0.41	(0.24-0.70)	.00
Need			
Age	1.01	(0.99-1.04)	.37
Pack-year smoking history, pack-years Smoking status	1.00	(0.99-1.01)	.79
Ref: formerly smoked	-	-	-
Currently smoke	0.64	(0.34-1.20)	.17
Predisposing			
Chronic obstructive pulmonary disease (COPD) diagnosis	2.62	(1.63-4.19)	.00
Number of chronic diseases	1.03	(0.95-1.12)	.49
Family history of lung cancer			
Ref: no family history	-	-	-
At least one first degree relative	1.56	(0.96-2.55)	.08
At least one second degree relative	0.87	(0.51-1.49)	.61
Gender			
Ref: Men	-	-	-
Women	0.44	(0.30-0.63)	.00
Race/Ethnicity			
Ref: White American	-	-	-
Black/African American	1.05	(0.73-1.51)	.81
Asian/Hispanic/Latino/Multiple Ethnicities	0.45	(0.15-1.40)	.17
Education level			
Ref: High School Equivalent or Less	-	-	-

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Variable Name	OR	95% CI	*P*-Value
Some college or associates	1.25	(0.81-1.93)	.31
Bachelors	1.09	(0.61-1.92)	.78
Postgraduate	2.02	(0.91,4.47)	.08
Marital status			
Ref: never married	-	-	-
Married	1.28	(0.78-2.10)	.32
No longer with spouse	1.04	(0.62-1.72)	.89
U.S. census region			
Ref: South	-	-	-
All other regions (Northeast-Midwest, West)	1.08	(0.75-1.54)	.69

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Table 3

Adjusted Associations Between Predisposing, Enabling, and Need Factors and Odds of Lung Cancer Screening Discussion[a]

Variable Name	OR	95% CI	*P*-Value
Enabling			
Health insurance status			
Ref: has insurance	-	-	-
No insurance	0.72	(0.29-1.79)	.48
Income			
Ref: Less than $30,000	-	-	-
$30,000-$59,999	1.78	(1.06-2.98)	.03
$60,000 or more	1.79	(0.93-3.45)	.08
Employment status			
Ref: employed	-	-	-
Unemployed/Other	1.18	(0.65-2.14)	.59
Retired	1.05	(0.58-1.90)	.88
Primary care provider (PCP) status			
Ref: has a PCP	-	-	-
Lack of a PCP	0.39	(0.21-0.71)	.00
Need			
Age	1.02	(0.98-1.06)	.37
Pack-year Smoking History-pack-years	0.99	(0.99-1.00)	.23
Smoking status			
Ref: formerly smoked	-	-	-
Currently smoke	0.38	(0.18-0.79)	.01
Predisposing			
Chronic obstructive pulmonary disease (COPD) diagnosis	3.56	(1.90-6.67)	.00
Number of chronic diseases	0.90	(0.81-1.02)	.09
Family history of lung cancer			
Ref: no family history	-	-	-
At least one first degree relative	1.78	(1.02-3.09)	.04
At least one second degree relative	1.06	(0.59-1.92)	.84
Gender			
Ref: Men	-	-	-
Women	0.37	(0.24-0.58)	.00
Race/Ethnicity			
Ref: White	-	-	-
Black/African American	1.17	(0.75-1.84)	.49
Asian/Hispanic/Latino/Multiple Ethnicities	0.75	(0.22-2.50)	.64
Education Level			

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Variable Name	OR	95% CI	*P*-Value
Ref: high school equivalent or less	-	-	-
Some college or associates	1.13	(0.69-1.85)	.62
Bachelors	1.02	(0.53-1.95)	.96
Postgraduate	1.72	(0.69-4.28)	.24
Marital status			
Ref: never married	-	-	-
Married	0.83	(0.45-1.53)	.54
No longer with spouse	1.01	(0.56-1.82)	.98
U.S. census region			
Ref: South	-	-	-
All other regions (Northeast, Midwest, West)	1.02	(0.68-1.55)	.91

[a]N = 481 for adjusted model due to missing data.